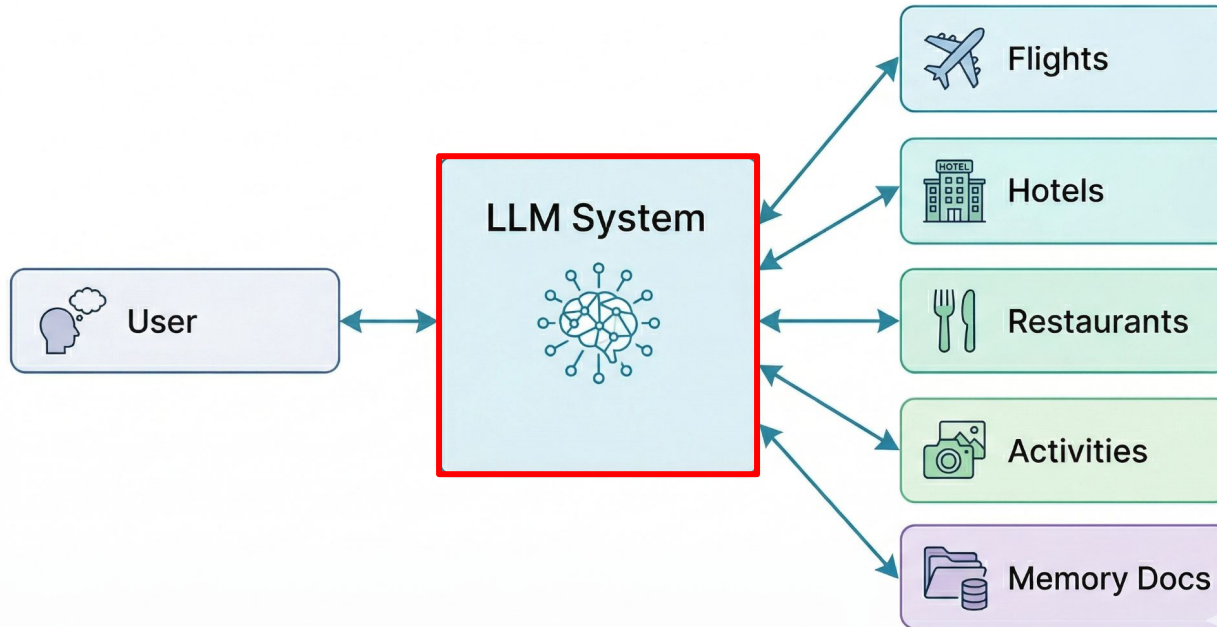




Final Project: Dynamic Travel Replanning

Project Overview

- Students will build a **LLM system** that replans travel under changing constraints.

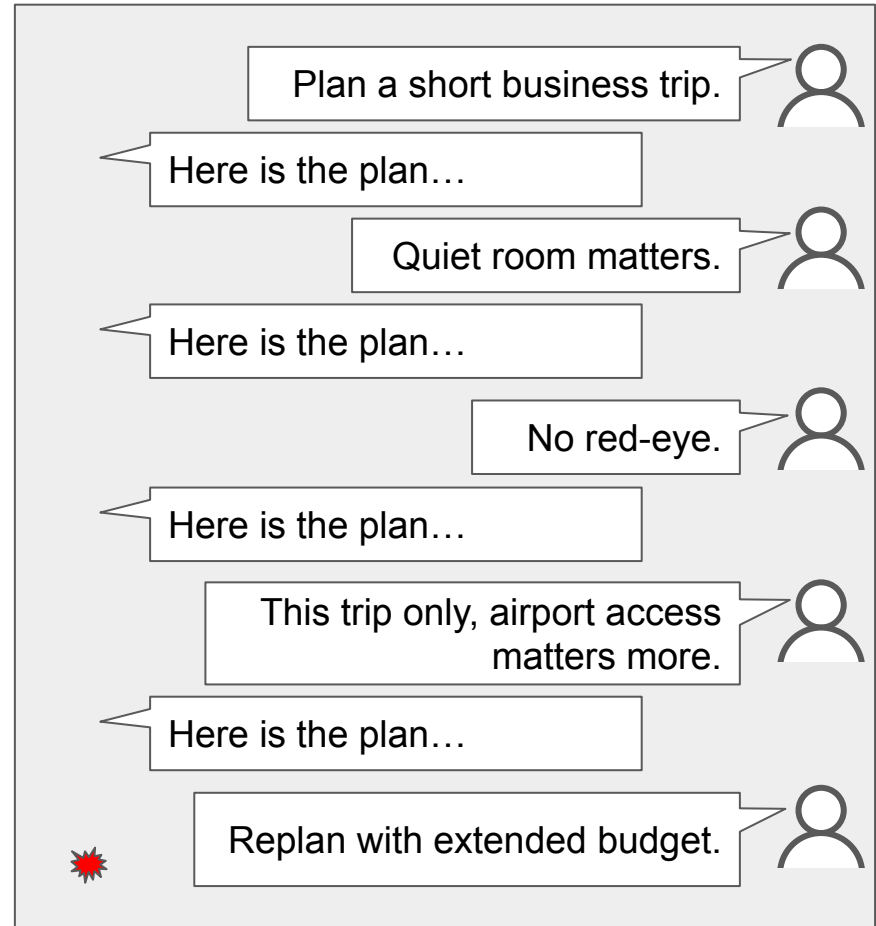


Problem Scenario

The user gives:

- an initial request
- follow-up constraints
- spoken rules
- updates that invalidate older assumptions

The system must produce a coherent final plan **while managing memory correctly.**



Example of Task

Each benchmark task is a **multi-turn episode**.

Example User Turn

- “Please build a short trip plan that keeps me functional for work, not just technically on time.”
- “Bring back only the relevant old preferences, not the whole history; keep active context lean.”
- “For this trip only, airport access matters more than local character.”
- “Count bundles only if they improve the full plan, not just the sticker price.”
- “Do not assume a nominal dinner bundle still works after a late arrival.”
- “The older budget assumption is no longer valid, so retire it.”

This task requires:

- Retrieve only the relevant memory and profile documents
- Apply the trip-only airport-access override
- Retire the stale budget assumption
- Evaluate bundle value at the full-plan level
- Reject dinner bundles that no longer work after a late arrival
- Return a grounded final bundle with valid IDs

Data and Tools for the Benchmark

- We offer data and primitive tools for the benchmark.
- You can use them as it is or improve tool usability by implementing wrapper tools
 - e.g., semantic_hotel_reranker, risk_aware_plan_verifier

Public data files

inventory_flights / hotels / restaurants / activities
profile_briefs
venue_briefs
city_ops_notes
memory_corpus
rejected_options_memory
partner_promotions
booking_constraints
option_dependencies
event_calendar

Primitive tools

search_memory
get_rejected_options
get_profile_brief
get_venue_brief
get_city_ops_notes
get_policy
search_flights
search_hotels
search_restaurants
search_activities

Baseline

- We included an example solver as the baseline
 - Single-agent planner
 - Queries profile, venue, policy, and memory
 - Searches primitive inventory tools
 - Submits grounded IDs + memory report
- Solver configuration
 - model: gpt-5.4-nano
 - embedding: text-embedding-3-small
 - max_tool_rounds: 9
 - max_tool_results: 4
 - max_output_tokens: 800

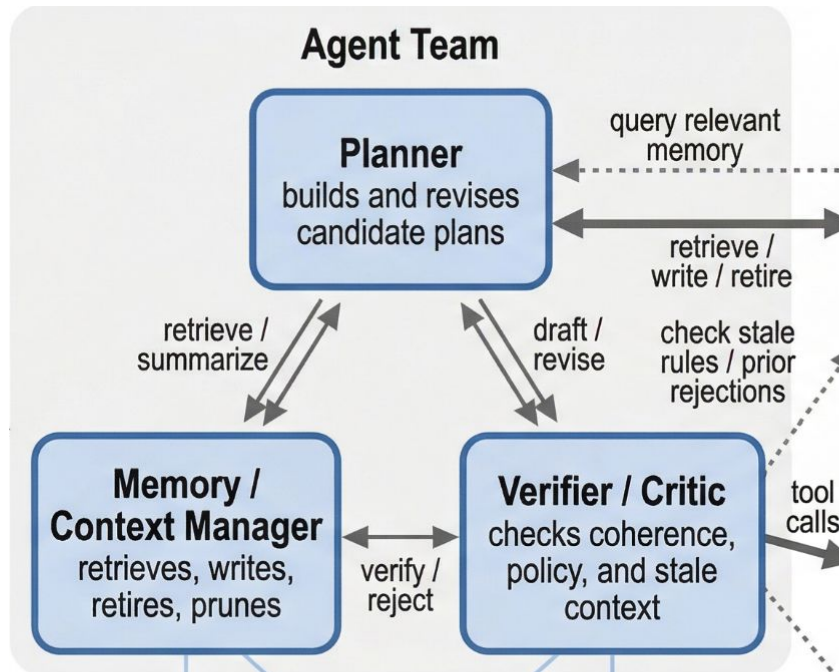
What to Implement?

Students may build an **agent team** include roles such as:

- a **Planner**
- a **Memory / Context Manager**
- a **Verifier / Critic**

The system should decide:

- what to retrieve
- what to remember
- what to retire
- what to keep in active context
- how agents communicate
- how to revise a weak plan



Implementation Flexibility and Fixed Specs

You are free to design

- Single agent vs. MAS
- Number of agents and role split
- Prompt design
- Memory write / retrieve / retire policy
- Rerankers, verifiers, and wrapper tools
- Search strategy
- Model parameters

TA-controlled and fixed

- Available models
- Tools and initial data
- Official runtime wrapper
- Usage and token-cost measurement
- Evaluator
- Hidden tasks and hidden gold
- Final grading run

Evaluation Set

The benchmark contains **50 tasks** in total:

- **Public:** 20 tasks
 - Easy **2**, Medium **7**, Hard **11**
- **Hidden:** 30 tasks
 - Easy **4**, Medium **13**, Hard **13**
- The benchmark emphasizes **context management under change**, including:
 - preference updates
 - stale-memory retirement
 - rejected-option persistence
 - multi-constraint tradeoffs
 - grounded replanning

Score Evaluation

40%

Feasibility / constraints

- hard constraint
- policy-safe
- valid bundle
- meeting-safe arrival

30%

Preference fit / plan quality

- decision quality
- semantic fit
- spoken-rule compliance

20%

Adaptation / memory discipline

- update handling
- memory retrieval / retirement
- distractor avoidance
- lean active context

10%

Efficiency bonus

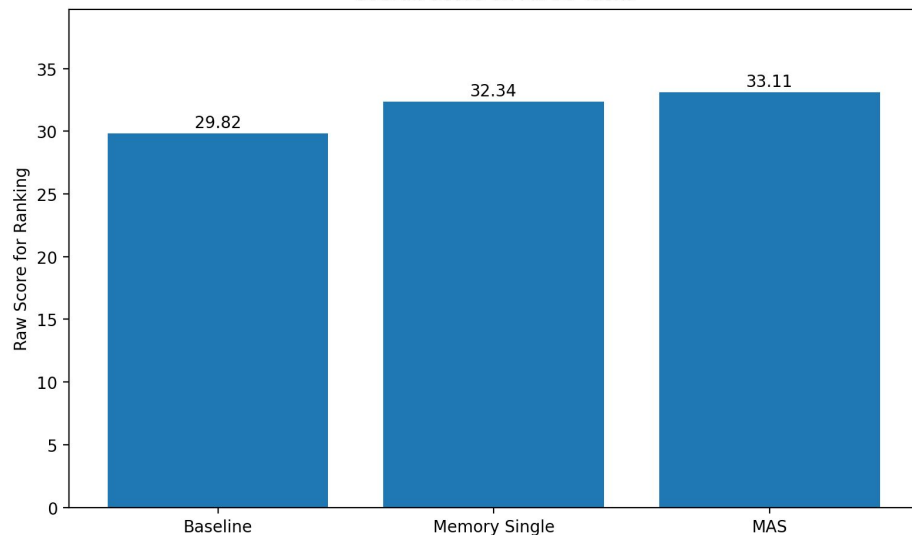
- lower cost
- but efficiency cannot rescue a bad plan

- We will rerun your submitted solver with the official runtime wrapper and evaluator on public 20 + hidden 30 episodes.
- Please use public 20 score as a development signal

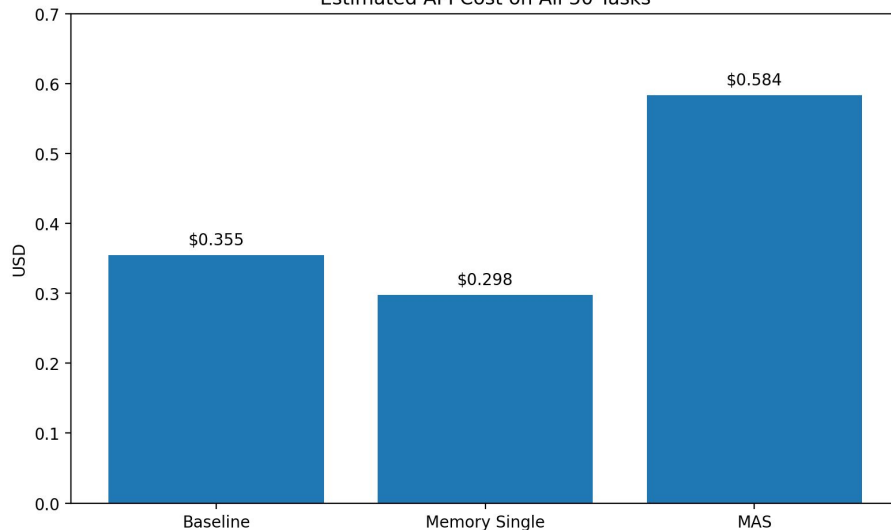
Test Results

- We evaluated score and estimated API cost on 50 tasks
 - Agent scheme: Baseline without memory, single agent with memory, multi-agent system

Overall Score on All 50 Tasks



Estimated API Cost on All 50 Tasks



Leaderboard (Tentative)

- We plan to provide a course leaderboard so teams can track progress during the project.
- Some details are still being finalized, including how submissions will be collected and how often the leaderboard will be updated.
 - Our current expectation is that your submitted solver will be re-run by the course staff in an official evaluation environment.
 - The leaderboard will likely consider both benchmark performance and efficiency.
- The final project score is expected to be based on your ranking on the final submission.
- We will share the final rules once the evaluation setup is fully confirmed.

Important Dates

- Final Project Status Check
 - We will check your initial status of the project.
 - F2F meeting, 20 minutes per each team.
 - Team 1~3: 2026-05-07 (Thu)
 - Team 4~6: 2026-05-12 (Tue)
- Deep Dive
 - F2F presentation to check individual contributions
 - Each team is required to book a 1-hour appointment within June 1 ~ 5.
 - https://docs.google.com/spreadsheets/d/1N4J7H0ElmkNC1fQpxNyJg7DXeQRZJXcNEz1NFiwo_dA/edit
- Final Presentation
 - Site: 942-302
 - Team 1~3: 2026-06-09 (Tue)
 - Team 4~6: 2026-06-11 (Thu)